CULTIVATION GUIDE | PESTS

Rice Pests in Sri Lanka: Recognition and Non-Chemical Management

The insects RRDI treats as major rice pests, with the signs, risk periods and cultural controls it publishes for the two most damaging.

Document AV-CG-08 | Category: cultivation_guide | RRDI pest pages, checked 3 Oct 2026
Teaching edition for the STEMLink AI Engineer Bootcamp, compiled from official Sri Lankan sources (Department of Agriculture and RRDI web pages, Agricultural and Agrarian Insurance Board, Cabinet decisions as reported), accessed 3 October 2026. Sources are listed in the corpus README, not in this document.

SCOPE

This guide describes pests so that a farmer's report can be understood and routed. AgriVibe does not diagnose pests and never names a pesticide or recommends a chemical treatment (AV-POL-01). Chemical control decisions belong to the Agriculture Instructor or the 1920 service.

1. Major rice pests

Based on the extent and severity of damage, RRDI lists the following as major pests of rice in Sri Lanka: thrips, brown planthopper, yellow stem borer, rice leaf-folders, rice sheath mite, rice gall midge and paddy bug. Grain moth, grain weevil and red flour beetle are pests of stored paddy and rice.

2. Brown planthopper (BPH)

Signs

Heavy infestations produce hopper burn: leaves dry and turn brown after feeding, and patches of burned plants are often lodged. BPH also carries the grassy stunt and ragged stunt virus diseases. The plant is most sensitive at the late vegetative and reproductive stages.

Economic threshold

Two insects per hill at booting, and five per hill at heading. Spiders are major predators of BPH, so the threshold is adjusted according to how many spiders are present.

Resistant varieties

Varieties with moderate resistance include Bg 379-2, Bg 300, Bg 403, Bg 304, Bg 357, Bg 358 and Bg 360.

Cultural management

- Cultivate resistant varieties.
- Draining the field to reduce moisture helps prevent BPH multiplication.
- Indiscriminate insecticide use during the vegetative stage is known to cause BPH outbreaks, because it kills natural enemies.
- Monitor the crop regularly for early signs of infestation.

3. Rice gall midge

Signs

The larvae feed inside the growing bud, which forms a gall called a silver shoot or onion shoot. A gall ends the tiller's development and so reduces yield. Severity depends on the growth stage at which the crop is attacked.

When damage is high

Damage is high in wet, humid weather. Infestation is high in the Dry and Intermediate Zones during Maha, and in the Wet Zone during Yala. A biotype detected after 1986 is called biotype II.

Resistant varieties

Resistant to biotype I: Bg 276-5, Bg 400-1, Bg 380, Bg 450 and Bg 300. Resistant to biotypes I and II: Bg 304, Bg 305, Bg 357, Bg 359 and Bg 360.

Cultural management

- Growing resistant varieties in areas where the pest is endemic is the most economical method.
- Control weeds.
- Chemical control applied after damage is seen gives poor results; prevention through variety choice matters more.